

T3 UV one-loop RGE term comparison

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \rightarrow g$ conversion as in the previous report.

$\beta_{g_Y}^{(1)}$

$$16\pi^2\,\mu\frac{dg_Y}{d\mu} = \beta_{g_Y}^{(1)}$$

Column Representative term structure	
T1	$\frac{43g_Y^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{43g_Y^3}{6}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{43g_Y^3}{6}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{59g_Y^3}{6}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{59g_Y^3}{6}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{59g_Y^3}{6}$

$\beta_{g_2}^{(1)}$

$16\pi^2\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$

Column Representative term structure	
T1	$-\frac{17g_2^3}{6}$
T2	$-\frac{g_2^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17g_2^3}{6}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{7g_2^3}{6}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$-\frac{g_2^3}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{g_2^3}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\frac{g_2^3}{6}$

$$\beta_{g_3}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column Representative term structure

T1	$-7g_3^3$
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-7g_3^3$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-7g_3^3$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-7g_3^3$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$

$\beta_{Y_u}^{(1)}$

$16\pi^2\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$
T2	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$
T3	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$

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Column	Representative term structure	
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	
T3	$-8g_3^2\ (Y_u)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Column	Representative term structure
T1	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

$16\pi^2\,\mu\,\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

$16\pi^2\,\mu\,\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$
T3	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Terms 2 of 3

Column	Representative term structure	
T1	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$	
T2	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	
T3	$-\frac{15}{4}g_Y^2\ (Y_e)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_Y^2\ (Y_e)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_Y^2\ (Y_e)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_Y^2\ (Y_e)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_Y^2\ (Y_e)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_Y^2\ (Y_e)_{ij}$

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Column	Representative term structure
T1	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{y_1}^{(1)}$

$16\pi^2\,\mu\,\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$

Terms 1 of 5

Column	Representative term structure	
T1	$\text{Tr}\Big(y_1\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	
T2	$+\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)\,(y_1)_{ij}$	
T3	$+\frac{1}{2}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij}+\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\Big(y_1\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)\,(y_1)_{ij}$	$+\frac{1}{2}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij}+\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)\,(y_1)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)\,(y_1)_{ij}$	—

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{9}{4}g_2^2(y_1)_{ij}$
T2	$-\frac{3}{4}g_Y^2(y_1)_{ij}$
T3	$\frac{3}{2}\text{Tr}\left(y_1(y_2)^\dagger\right)(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9}{4}g_2^2(y_1)_{ij}$	$-\frac{3}{4}g_Y^2(y_1)_{ij}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{33}{4}g_2^2(y_1)_{ij}$	$-\frac{3}{4}g_Y^2(y_1)_{ij}$	$\frac{3}{2}\text{Tr}\left(y_1(y_2)^\dagger\right)(y_2)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_1)_{ij}$	$-\frac{3}{2}g_Y^2(y_1)_{ij}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_1)_{ij}$	$-\frac{3}{2}g_Y^2(y_1)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_1)_{ij}$	$-\frac{3}{2}g_Y^2(y_1)_{ij}$	—

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Column	Representative term structure
T1	$+\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$
T2	$+\frac{1}{4}\left(2\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij}\right)$
T3	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$	$+\frac{1}{4}\left(2\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij}\right)$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure	
T1	$+ \frac{3}{4} \left(y_2 \left(y_2 \right)^{\dagger} y_1 \right)_{ij}$	
T2	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_1 \right)_{ij}$	
T3	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \frac{3}{4} \left(y_2 \left(y_2 \right)^{\dagger} y_1 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_1 \right)_{ij}$	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+\frac{1}{2}((y_2\ (y_2)^\dagger\ y_1)_{ij} + (Y_e\ (Y_e)^\dagger\ y_1)_{ij} + 3\ (y_1\ (y_1)^\dagger\ y_1)_{ij})$
T2	$+\frac{1}{4}(3\ (y_2\ (y_2)^\dagger\ y_1)_{ij} + 2\ (Y_e\ (Y_e)^\dagger\ y_1)_{ij} + 6\ (y_1\ (y_1)^\dagger\ y_1)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{2}((y_2\ (y_2)^\dagger\ y_1)_{ij} + (Y_e\ (Y_e)^\dagger\ y_1)_{ij} + 3\ (y_1\ (y_1)^\dagger\ y_1)_{ij})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{1}{4}(3\ (y_2\ (y_2)^\dagger\ y_1)_{ij} + 2\ (Y_e\ (Y_e)^\dagger\ y_1)_{ij} + 6\ (y_1\ (y_1)^\dagger\ y_1)_{ij})$

$\beta_{y_2}^{(1)}$

$16\pi^2\mu\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$

Terms 1 of 5

Column	Representative term structure	
T1	$\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)\,(y_1)_{ij}$	
T2	$+\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	
T3	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+2\,(y_2\ (y_1)^\dagger\ y_1)_{ij}+(\textcolor{teal}{Y}_e\ (\textcolor{teal}{Y}_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)\,(y_1)_{ij}$	$+\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+2\,(y_2\ (y_1)^\dagger\ y_1)_{ij}+(\textcolor{teal}{Y}_e\ (\textcolor{teal}{Y}_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	—

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{9}{4}g_2^2(y_2)_{ij}$
T2	$-\frac{3}{4}g_Y^2(y_2)_{ij}$
T3	$\frac{3}{2}\text{Tr}\left(y_2(y_1)^\dagger\right)(y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9}{4}g_2^2(y_2)_{ij}$	$-\frac{3}{4}g_Y^2(y_2)_{ij}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{33}{4}g_2^2(y_2)_{ij}$	$-\frac{3}{4}g_Y^2(y_2)_{ij}$	$\frac{3}{2}\text{Tr}\left(y_2(y_1)^\dagger\right)(y_1)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_2)_{ij}$	$-\frac{3}{2}g_Y^2(y_2)_{ij}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_2)_{ij}$	$-\frac{3}{2}g_Y^2(y_2)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_2)_{ij}$	$-\frac{3}{2}g_Y^2(y_2)_{ij}$	—

Terms 3 of 5

Column	Representative term structure
T1	$+\frac{3}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$
T2	$+\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$
T3	$+\frac{1}{2}\left(\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$	$+\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+\frac{1}{2}\left(\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$2 \operatorname{Tr} \left(y_2 \left(y_2 \right)^{\dagger} \right) \left(y_2 \right)_{ij}$
T2	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$
T3	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$2 \operatorname{Tr} \left(y_2 \left(y_2 \right)^{\dagger} \right) \left(y_2 \right)_{ij}$	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+ \left(y_2 \ (y_2)^\dagger \ y_2\right)_{ij}$
T2	$+ \frac{1}{4} (3 \left(y_1 \ (y_1)^\dagger \ y_2\right)_{ij} + 2 \left(Y_e \ (Y_e)^\dagger \ y_2\right)_{ij} + 6 \left(y_2 \ (y_2)^\dagger \ y_2\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \left(y_2 \ (y_2)^\dagger \ y_2\right)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+ \frac{1}{4} (3 \left(y_1 \ (y_1)^\dagger \ y_2\right)_{ij} + 2 \left(Y_e \ (Y_e)^\dagger \ y_2\right)_{ij} + 6 \left(y_2 \ (y_2)^\dagger \ y_2\right)_{ij})$

$\beta_{M_F}^{(1)}$

$16\pi^2\mu\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}$
T2	$+ \left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$
T3	$+ \left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}$	$+ \left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$	$+ \left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$+ \left((y_2)^T \, (y_2)^* \, M_F \right)_{ij}$
T2	$\frac{1}{2} \left(\left((M_F)^T \, (y_1)^\dagger \, y_1 \right)_{ij} + \left((y_1)^T \, (y_1)^* \, M_F \right)_{ij} + \left((M_F)^T \, (y_2)^\dagger \, y_2 \right)_{ij} + \left((y_2)^T \, (y_2)^* \, M_F \right)_{ij} \right)$
T3	$- 6 g_2^2 \left(\left((M_F)^T \right)_{ij} + (M_F)_{ij} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \left((y_2)^T \, (y_2)^* \, M_F \right)_{ij}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{1}{2} \left(\left((M_F)^T \, (y_1)^\dagger \, y_1 \right)_{ij} + \left((y_1)^T \, (y_1)^* \, M_F \right)_{ij} + \left((M_F)^T \, (y_2)^\dagger \, y_2 \right)_{ij} + \left((y_2)^T \, (y_2)^* \, M_F \right)_{ij} \right)$	$- 6 g_2^2 \left(\left((M_F)^T \right)_{ij} + (M_F)_{ij} \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$\frac{1}{4}(2\left((y_1)^T\left(y_1\right)^{*} M_F\right)_{i j}+3\left(M_F\left(y_2\right)^{\dagger} y_2\right)_{i j})$
T2	$-\frac{9}{2} g_2^2\left(M_F\right)_{i j}$
T3	$-\frac{3}{2} g_Y^2\left(M_F\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{1}{4}(2\left((y_1)^T\left(y_1\right)^{*} M_F\right)_{i j}+3\left(M_F\left(y_2\right)^{\dagger} y_2\right)_{i j})-\frac{9}{2} g_2^2\left(M_F\right)_{i j}$	$-\frac{9}{2} g_2^2\left(M_F\right)_{i j}$	$-\frac{3}{2} g_Y^2\left(M_F\right)_{i j}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{1}{4}(3\left((y_1)^T\left(y_1\right)^{*} M_F\right)_{i j}+2\left(M_F\left(y_2\right)^{\dagger} y_2\right)_{i j})-\frac{9}{2} g_2^2\left(M_F\right)_{i j}$	$-\frac{9}{2} g_2^2\left(M_F\right)_{i j}$	$-\frac{3}{2} g_Y^2\left(M_F\right)_{i j}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{2} g_2^2\left(M_F\right)_{i j}$	$-\frac{3}{2} g_Y^2\left(M_F\right)_{i j}$

Terms 4 of 4

Column	Representative term structure
T1	$\frac{3}{4}((y_1)^T \ (y_1)^* \ M_F)_{ij} + (M_F \ (y_2)^\dagger \ y_2)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3}{4}((y_1)^T \ (y_1)^* \ M_F)_{ij} + (M_F \ (y_2)^\dagger \ y_2)_{ij})$

$\beta_{m_{S_1}^2}^{(1)}$

$16\pi^2\mu\frac{dm_{S_1}^2}{d\mu}=\beta_{m_{S_1}^2}^{(1)}$

Terms 1 of 4

Column		Representative term structure
T1	$m_{S_1}^2$	$(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+6\lambda_{S_1})$
T2	$-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$	
T3	$-8\text{Tr}\left(M_F\left(y_1\right)^{\dagger}y_1\left(M_F\right)^*\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$m_{S_1}^2(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+6\lambda_{S_1})$	$-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$	$-8\text{Tr}\left(M_F\left(y_1\right)^{\dagger}y_1\left(M_F\right)^*\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$m_{S_1}^2(3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+6\lambda_{S_1})$	$-12\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$	$-12\text{Tr}\left(M_F\left(y_1\right)^{\dagger}y_1\left(M_F\right)^*\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$m_{S_1}^2(4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+4\lambda_{S_1})$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{9g_2^2m_{S_1}^2}{2}$
T2	$-\frac{3g_Y^2m_{S_1}^2}{2}$
T3	$+4\lambda_{12}m_{S_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2m_{S_1}^2}{2}$	$-\frac{3g_Y^2m_{S_1}^2}{2}$	$+4\lambda_{12}m_{S_2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9g_2^2m_{S_1}^2}{2}$	$-\frac{3g_Y^2m_{S_1}^2}{2}$	$+4\lambda_{12}m_{S_2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+6\lambda_{12}m_{S_2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+2\lambda_{12}m_{S_2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$-8\text{Tr}\left(M_F\left(M_F\right)^{\dagger}\left(y_1\right)^T\left(y_1\right)^{*}\right)$
T2	$m_{S_1}^2\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}}\right)$
T3	$-12g_2^2m_{S_1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-8\text{Tr}\left(M_F\left(M_F\right)^{\dagger}\left(y_1\right)^T\left(y_1\right)^{*}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-4\text{Tr}\left(M_F\left(M_F\right)^{\dagger}\left(y_1\right)^T\left(y_1\right)^{*}\right)$	$m_{S_1}^2\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}}\right)$	$-12g_2^2m_{S_1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-4\text{Tr}\left(M_F\left(M_F\right)^{\dagger}\left(y_1\right)^T\left(y_1\right)^{*}\right)$	$m_{S_1}^2\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}}\right)$	$-12g_2^2m_{S_1}^2$

Terms 4 of 4

Column	Representative term structure
T1	$-6g_Y{}^2m_{S_1}^2$
T2	$+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-6g_Y{}^2m_{S_1}^2$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{m_{S_2}^2}^{(1)}$

$$16\pi^2\,\mu\,\frac{dm_{S_2}^2}{d\mu}=\beta_{m_{S_2}^2}^{(1)}$$

Terms 1 of 4

Column		Representative term structure
T1	$m_{S_2}^2$	$(2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+6\lambda_{S_2})$
T2	$-8\,\text{Tr}\big(M_F\,(M_F)^*\,(y_2)^T\,(y_2)^*\big)$	
T3	$-8\,\text{Tr}\big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^*\big)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$m_{S_2}^2\,(2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+6\lambda_{S_2})$	$-8\,\text{Tr}\big(M_F\,(M_F)^*\,(y_2)^T\,(y_2)^*\big)$	$-8\,\text{Tr}\big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^*\big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$m_{S_2}^2\,(3\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+6\lambda_{S_2})$	$-12\,\text{Tr}\big(M_F\,(M_F)^*\,(y_2)^T\,(y_2)^*\big)$	$-12\,\text{Tr}\big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^*\big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$m_{S_2}^2\,(4\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+4\lambda_{S_2})$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{9g_2^2m_{S_2}^2}{2}$
T2	$-\frac{3g_Y^2m_{S_2}^2}{2}$
T3	$+4\lambda_{12}m_{S_1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2m_{S_2}^2}{2}$	$-\frac{3g_Y^2m_{S_2}^2}{2}$	$+4\lambda_{12}m_{S_1}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9g_2^2m_{S_2}^2}{2}$	$-\frac{3g_Y^2m_{S_2}^2}{2}$	$+4\lambda_{12}m_{S_1}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_{12}m_{S_1}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+6\lambda_{12}m_{S_1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$m_{S_2}^2 (2 \operatorname{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + 8 \lambda_{S_2} + 4 \lambda_{S_2}^{\text{Adj}})$
T2	$-4 \operatorname{Tr} \left(M_F \left(y_2 \right)^\dagger y_2 \left(M_F \right)^\dagger \right)$
T3	$-12 g_2^2 m_{S_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$m_{S_2}^2 (2 \operatorname{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + 8 \lambda_{S_2} + 4 \lambda_{S_2}^{\text{Adj}})$	$-4 \operatorname{Tr} \left(M_F \left(y_2 \right)^\dagger y_2 \left(M_F \right)^\dagger \right)$	$-12 g_2^2 m_{S_2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-8 \operatorname{Tr} \left(M_F \left(y_2 \right)^\dagger y_2 \left(M_F \right)^\dagger \right)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$m_{S_2}^2 (2 \operatorname{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + 8 \lambda_{S_2} + 4 \lambda_{S_2}^{\text{Adj}})$	$-4 \operatorname{Tr} \left(M_F \left(y_2 \right)^\dagger y_2 \left(M_F \right)^\dagger \right)$	$-12 g_2^2 m_{S_2}^2$

Terms 4 of 4

Column	Representative term structure
T1	$-6g_Y^2 m_{S_2}^2$
T2	$+m_{S_1}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-6g_Y^2 m_{S_2}^2$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-6g_Y^2 m_{S_2}^2$	$+m_{S_1}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_H}^{(1)}$

$16\pi^2\mu\frac{d\lambda_H}{d\mu}=\beta_{\lambda_H}^{(1)}$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$
T2	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$
T3	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$

Terms 2 of 7

Column	Representative term structure
T1	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$
T2	$+8 \lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\left(\lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\right)^{*}$
T3	$+4 \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\left(\lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+8 \lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\left(\lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\right)^{*}$	$+4 \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\left(\lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\right)^{*}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+8 \lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\left(\lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\right)^{*}$	$+4 \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\left(\lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\right)^{*}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	—	—

Terms 3 of 7

Column	Representative term structure
T1	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$
T2	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$

Terms 4 of 7

Column	Representative term structure
T1	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$
T2	$-9g_2^2\lambda_H$
T3	$+\frac{3g_Y^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$

Terms 5 of 7

Column	Representative term structure
T1	$+12\lambda_H^2$
T2	$+4\lambda_{H1}^2$
T3	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+2\lambda_{H1}^2$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	—

Terms 6 of 7

Column	Representative term structure
T1	$+4\lambda_{H2}^2$
T2	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3	$+3\lambda_{H2}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H2}^2$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{H2}^2$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{H2}^2$	—	$+3\lambda_{H2}^{\text{Adj}^2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{H2}^2$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{H2}^2$	—	$+3\lambda_{H2}^{\text{Adj}^2}$

Column	Representative term structure
T1	$+3\lambda_{H1}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$

$\beta_{\lambda_{S_1}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}}{d\mu}=\beta_{\lambda_{S_1}}^{(1)}$

Terms 1 of 9

Column	Representative term structure
T1	$4\lambda_{S_1}\text{Tr}\big(y_1\ (y_1)^\dagger\big)$
T2	$+\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}(2\text{Tr}\big(y_1\ (y_2)^\dagger\big)+24\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*)$
T3	$+2\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_{S_1}\text{Tr}\big(y_1\ (y_1)^\dagger\big)$	$+\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}(2\text{Tr}\big(y_1\ (y_2)^\dagger\big)+24\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*)+2\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$6\lambda_{S_1}\text{Tr}\big(y_1\ (y_1)^\dagger\big)$	$+\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}(3\text{Tr}\big(y_1\ (y_2)^\dagger\big)+24\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*)+3\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$8\lambda_{S_1}\text{Tr}\big(y_1\ (y_1)^\dagger\big)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 9

Column	Representative term structure
T1	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T2	$+8\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$
T3	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+8\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-5\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+8\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-8\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—	—

Terms 3 of 9

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$
T3	$-9g_2^2 \lambda_{S_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{9g_2^4}{4}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$	$-9g_2^2 \lambda_{S_1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{9g_2^4}{4}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$	$-9g_2^2 \lambda_{S_1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 9

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{4}$
T2	$+4\lambda_{12}^2$
T3	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{3g_Y^4}{4}$	$+4\lambda_{12}^2$	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{3g_Y^4}{4}$	$+4\lambda_{12}^2$	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^2$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+2\lambda_{12}^2$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^2$	—

Terms 5 of 9

Column	Representative term structure
T1	$+4\lambda_{H1}^2$
T2	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$
T3	$+12\lambda_{S1}^2$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H1}^2$	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$	$+12\lambda_{S1}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{H1}^2$	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$	$+12\lambda_{S1}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H1}^2$	—	$+10\lambda_{S1}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{H1}^2$	—	$+14\lambda_{S1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H1}^2$	—	$+14\lambda_{S1}^2$

Terms 6 of 9

Column	Representative term structure
T1	$\lambda_{S_1} (4 \operatorname{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$
T2	$+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} (y_1 (y_1)^\dagger)$
T3	$+21 g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{S_1} (4 \operatorname{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_1} (4 \operatorname{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$

Terms 7 of 9

Column	Representative term structure
T1	$+g_Y^2\left(12g_2^2-12\lambda_{S_1}-6\lambda_{S_1}^{\text{Adj}}\right)$
T2	$+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$
T3	$+12g_Y^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_Y^2\left(12g_2^2-12\lambda_{S_1}-6\lambda_{S_1}^{\text{Adj}}\right)$	$+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$	$+12g_Y^4$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$	—

Terms 8 of 9

Column	Representative term structure
T1	$+ \frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$
T2	$+26\lambda_{S1}^{\text{Adj}^2}$
T3	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$	$+26\lambda_{S1}^{\text{Adj}^2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$	$+26\lambda_{S1}^{\text{Adj}^2}$	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$

Terms 9 of 9

Column	Representative term structure
T1	$+6\lambda_{12}^{\text{Adj}^2}$
T2	$+6\lambda_{12}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$

$\beta_{\lambda_{S_2}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_2}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$

Terms 1 of 9

Column Representative term structure	
T1	$\lambda_{S_1^\dagger S_2 S_2^\dagger 2}(2\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+24\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\text{)}$
T2	$+2\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$
T3	$+4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{S_1^\dagger S_2 S_2^\dagger 2}(2\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+24\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\text{)}$	$+2\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{S_1^\dagger S_2 S_2^\dagger 2}(3\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+24\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\text{)}$	$+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 9

Column	Representative term structure
T1	$-4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T2	$+8\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$
T3	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	$+8\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-5\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	$+8\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-8\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—

Terms 3 of 9

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$
T3	$-9g_2^2 \lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$	$-9g_2^2 \lambda_{S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{9g_2^4}{4}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$	$-9g_2^2 \lambda_{S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 9

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{4}$
T2	$+4\lambda_{12}^2$
T3	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{3g_Y^4}{4}$	$+4\lambda_{12}^2$	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{3g_Y^4}{4}$	$+4\lambda_{12}^2$	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+2\lambda_{12}^2$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+6\lambda_{12}^2$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^2$	—

Terms 5 of 9

Column	Representative term structure
T1	$+4\lambda_{H2}^2$
T2	$+\frac{\lambda_{H2}^{\text{Adj}2}}{4}$
T3	$+12\lambda_{S2}^2$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H2}^2$	$+\frac{\lambda_{H2}^{\text{Adj}2}}{4}$	$+12\lambda_{S2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{H2}^2$	$+\frac{\lambda_{H2}^{\text{Adj}2}}{4}$	$+12\lambda_{S2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+14\lambda_{S2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+10\lambda_{S2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+14\lambda_{S2}^2$

Terms 6 of 9

Column	Representative term structure
T1	$\lambda_{S_2} (4 \operatorname{Tr} (y_2 \ (y_2)^\dagger) + 14 \lambda_{S_2}^{\text{Adj}})$
T2	$+2 \lambda_{S_2}^{\text{Adj}} \operatorname{Tr} (y_2 \ (y_2)^\dagger)$
T3	$+21 g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2} (4 \operatorname{Tr} (y_2 \ (y_2)^\dagger) + 14 \lambda_{S_2}^{\text{Adj}})$	$+2 \lambda_{S_2}^{\text{Adj}} \operatorname{Tr} (y_2 \ (y_2)^\dagger)$	$+21 g_2^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2} (4 \operatorname{Tr} (y_2 \ (y_2)^\dagger) + 14 \lambda_{S_2}^{\text{Adj}})$	$+2 \lambda_{S_2}^{\text{Adj}} \operatorname{Tr} (y_2 \ (y_2)^\dagger)$	$+21 g_2^4$

Terms 7 of 9

Column	Representative term structure
T1	$+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$
T2	$+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$
T3	$+12g_Y^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2\left(12g_2{}^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2{}^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$	$+12g_Y{}^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2\left(12g_2{}^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2{}^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$	$+12g_Y{}^4$

Terms 8 of 9

Column	Representative term structure
T1	$+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$
T2	$+26\lambda_{S2}^{\text{Adj}^2}$
T3	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+26\lambda_{S2}^{\text{Adj}^2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+26\lambda_{S2}^{\text{Adj}^2}$	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$

Terms 9 of 9

Column	Representative term structure
T1	$+6\lambda_{12}^{\text{Adj}^2}$
T2	$+6\lambda_{12}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$

$$\beta_{\lambda_{H1}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{d\lambda_{H1}}{d\mu}=\beta_{\lambda_{H1}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure	
T1	$\lambda_{H1}(2\,\text{Tr}\big(y_1\,\,(y_1)^\dagger\big)+6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$	
T2	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\,\text{Tr}\big(y_1\,\,(y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)$	
T3	$+ \frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\,\text{Tr}\big(y_2\,\,(y_1)^\dagger\big)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H1}(2\,\text{Tr}\big(y_1\,\,(y_1)^\dagger\big)+6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\,\text{Tr}\big(y_1\,\,(y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)$	$+ \frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\,\text{Tr}\big(y_2\,\,(y_1)^\dagger\big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H1}(3\,\text{Tr}\big(y_1\,\,(y_1)^\dagger\big)+6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{4}\,\text{Tr}\big(y_1\,\,(y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)$	$+ \frac{3}{4}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\,\text{Tr}\big(y_2\,\,(y_1)^\dagger\big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H1}(4\,\text{Tr}\big(y_1\,\,(y_1)^\dagger\big)+6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 8

Column	Representative term structure
T1	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$
T2	$+24\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$
T3	$+3\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$+24\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$	$+3\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$+24\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$	$+3\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—

Terms 3 of 8

Column	Representative term structure
T1	$+6\lambda_{T3} \left(\lambda_{T3}\right)^*$
T2	$+\frac{9g_2^4}{4}$
T3	$-9g_2^2\lambda_{H1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{T3} \left(\lambda_{T3}\right)^*$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	—	—

Terms 4 of 8

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{4}$
T2	$-3g_Y^2 \lambda_{H1}$
T3	$+4\lambda_{12} \lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3g_Y^4}{4}$	$-3g_Y^2 \lambda_{H1}$	$+4\lambda_{12} \lambda_{H2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3g_Y^4}{4}$	$-3g_Y^2 \lambda_{H1}$	$+4\lambda_{12} \lambda_{H2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+6\lambda_{12} \lambda_{H2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+2\lambda_{12} \lambda_{H2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 8

Column	Representative term structure
T1	$+6\lambda_H\lambda_{H1}$
T2	$+4\lambda_{H1}^2$
T3	$+6\lambda_{H1}\lambda_{S1}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$	$+6\lambda_{H1}\lambda_{S1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$	$+6\lambda_{H1}\lambda_{S1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$	$+4\lambda_{H1}\lambda_{S1}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$	$+8\lambda_{H1}\lambda_{S1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$	$+8\lambda_{H1}\lambda_{S1}$

Terms 6 of 8

Column	Representative term structure
T1	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T2	$-\frac{9g_2^2\lambda_{H1}}{2}$
T3	$-\frac{3g_Y^2\lambda_{H1}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9g_2^2\lambda_{H1}}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{33g_2^2\lambda_{H1}}{2}$	$-\frac{15g_Y^2\lambda_{H1}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{33g_2^2\lambda_{H1}}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$

Terms 7 of 8

Column		Representative term structure
T1		$\lambda_{H1}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S1}^{\text{Adj}})$
T2		$+6g_2^4$
T3		$+3g_Y^4$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{H1}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S1}^{\text{Adj}})$	$+6g_2^4$	$+3g_Y^4$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H1}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S1}^{\text{Adj}})$	$+6g_2^4$	—

Terms 8 of 8

Column	Representative term structure
T1	$+6\lambda_{H1}^{\text{Adj}^2}$
T2	$+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_{H1}^{\text{Adj}^2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{H1}^{\text{Adj}^2}$	$+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$$\beta_{\lambda_{H2}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{d\lambda_{H2}}{d\mu}=\beta_{\lambda_{H2}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure
T1	$\lambda_{H2}(2\,\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\,\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\,\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^{\dagger 2}\right)^*\right)$
T3	$+ \frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\,\text{Tr}\big(y_2\ (y_1)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H2}(2\,\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\,\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\,\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^{\dagger 2}\right)^*\right)$	$+ \frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\,\text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H2}(3\,\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\,\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{4}\,\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^{\dagger 2}\right)^*\right)$	$+ \frac{3}{4}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\,\text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{H2}(4\,\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\,\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 8

Column	Representative term structure
T1	$-2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$
T2	$+3\lambda_{S_1^{\dagger}S_2S_2^{\dagger}2}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$
T3	$+24\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$+3\lambda_{S_1^{\dagger}S_2S_2^{\dagger}2}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$	$+24\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$+3\lambda_{S_1^{\dagger}S_2S_2^{\dagger}2}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$	$+24\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—

Terms 3 of 8

Column	Representative term structure
T1	$+6\lambda_{T3} \left(\lambda_{T3}\right)^*$
T2	$+\frac{9g_2^4}{4}$
T3	$-9g_2^2\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+12\lambda_{T3} \left(\lambda_{T3}\right)^*$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	—	—

Terms 4 of 8

Column	Representative term structure
T1	$+\frac{3g_Y^4}{4}$
T2	$-3g_Y^2\lambda_{H2}$
T3	$+4\lambda_{12}\lambda_{H1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H2}$	$+4\lambda_{12}\lambda_{H1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H2}$	$+4\lambda_{12}\lambda_{H1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_{12}\lambda_{H1}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+6\lambda_{12}\lambda_{H1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 8

Column	Representative term structure
T1	$+6\lambda_H\lambda_{H2}$
T2	$+4\lambda_{H2}^2$
T3	$+6\lambda_{H2}\lambda_{S2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$	$+6\lambda_{H2}\lambda_{S2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$	$+6\lambda_{H2}\lambda_{S2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$	$+8\lambda_{H2}\lambda_{S2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$	$+4\lambda_{H2}\lambda_{S2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$	$+8\lambda_{H2}\lambda_{S2}$

Terms 6 of 8

Column	Representative term structure
T1	$+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T2	$\lambda_{H2}(2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{S_2}^{\text{Adj}})$
T3	$+6g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{H2}(2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{S_2}^{\text{Adj}})$	$+6g_2^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{H2}(2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{S_2}^{\text{Adj}})$	$+6g_2^4$

Terms 7 of 8

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{H2}}{2}$
T2	$+3g_Y^4$
T3	$-\frac{15g_Y^2\lambda_{H2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{9g_2^2\lambda_{H2}}{2}$	—	$-\frac{3g_Y^2\lambda_{H2}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H2}}{2}$

Terms 8 of 8

Column	Representative term structure
T1	$+6\lambda_{H2}^{\text{Adj}^2}$
T2	$+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	$+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_{12}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$

Terms 1 of 11

Column	Representative term structure
T1	$\lambda_{12}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$
T2	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\frac{3}{2}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$
T3	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\left(\frac{3}{2}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\frac{3}{2}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\left(\frac{3}{2}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{12}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+3\text{Tr}\big(y_2\ (y_2)^\dagger\big))$	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\frac{9}{4}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\left(\frac{9}{4}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 11

Column	Representative term structure	
T1	$+\frac{3}{2}((\lambda_{S_1^\dagger S_2 S_2^\dagger 2})^* + (\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})^*) \text{Tr}(y_2 (y_1)^\dagger)$	
T2	$-2 \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$	
T3	$-4 \text{Tr}(y_1 (y_2)^\dagger y_2 (y_1)^\dagger)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}((\lambda_{S_1^\dagger S_2 S_2^\dagger 2})^* + (\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})^*) \text{Tr}(y_2 (y_1)^\dagger)$	$-2 \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$	$-4 \text{Tr}(y_1 (y_2)^\dagger y_2 (y_1)^\dagger)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{9}{4}((\lambda_{S_1^\dagger S_2 S_2^\dagger 2})^* + (\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})^*) \text{Tr}(y_2 (y_1)^\dagger)$	—	$-3 \text{Tr}(y_1 (y_2)^\dagger y_2 (y_1)^\dagger)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$-4 \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-4 \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-2 \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$	—

Terms 3 of 11

Column	Representative term structure
T1	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T2	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$
T3	$+3\lambda_{T3} (\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$	$+3\lambda_{T3} (\lambda_{T3})^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$	$+3\lambda_{T3} (\lambda_{T3})^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+4\lambda_{T3} (\lambda_{T3})^*$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+4\lambda_{T3} (\lambda_{T3})^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$-9g_2^2 \lambda_{12}$
T3	$+ \frac{3g_Y^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{9g_2^4}{4}$	$-9g_2^2 \lambda_{12}$	$+ \frac{3g_Y^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{9g_2^4}{4}$	$-9g_2^2 \lambda_{12}$	$+ \frac{3g_Y^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$-12g_2^2 \lambda_{12}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-12g_2^2 \lambda_{12}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 11

Column	Representative term structure
T1	$-3g_Y^2\lambda_{12}$
T2	$+4\lambda_{12}^2$
T3	$+6\lambda_{12}\lambda_{S_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$	$+6\lambda_{12}\lambda_{S_1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$	$+6\lambda_{12}\lambda_{S_1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$	$+4\lambda_{12}\lambda_{S_1}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$	$+8\lambda_{12}\lambda_{S_1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{12}^2$	—

Terms 6 of 11

Column	Representative term structure
T1	$+6\lambda_{12}\lambda_{S_2}$
T2	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3	$+4\lambda_{H1}\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{12}\lambda_{S_2}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$	$+4\lambda_{H1}\lambda_{H2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{12}\lambda_{S_2}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$	$+4\lambda_{H1}\lambda_{H2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{12}\lambda_{S_2}$	—	$+4\lambda_{H1}\lambda_{H2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{12}\lambda_{S_2}$	—	$+4\lambda_{H1}\lambda_{H2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+4\lambda_{H1}\lambda_{H2}$

Terms 7 of 11

Column	Representative term structure
T1	$-\frac{9}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T2	$\lambda_{12}(4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+4\lambda_{S_2}^{\text{Adj}})$
T3	$\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+4\lambda_{S_1}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{12}(4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+4\lambda_{S_2}^{\text{Adj}})$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+4\lambda_{S_1}^{\text{Adj}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 8 of 11

Column	Representative term structure	
T1	$\lambda_{12} \left(2 \operatorname{Tr} \left(y_1 \ (y_1)^\dagger \right) + 2 \operatorname{Tr} \left(y_2 \ (y_2)^\dagger \right) + \frac{16 \lambda_{12}^{\text{Cross}}}{3} + 4 \lambda_{S_1}^{\text{Adj}} + 4 \lambda_{S_2}^{\text{Adj}} \right)$	
T2	$+ \lambda_{12}^{\text{Cross}} \left(\frac{4}{3} \operatorname{Tr} \left(y_1 \ (y_1)^\dagger \right) + \frac{4}{3} \operatorname{Tr} \left(y_2 \ (y_2)^\dagger \right) \right)$	
T3	$+ \frac{4}{3} \lambda_{T3} \ (\lambda_{T3})^*$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{12} \left(2 \operatorname{Tr} \left(y_1 \ (y_1)^\dagger \right) + 2 \operatorname{Tr} \left(y_2 \ (y_2)^\dagger \right) + \frac{16 \lambda_{12}^{\text{Cross}}}{3} + 4 \lambda_{S_1}^{\text{Adj}} + 4 \lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(\frac{4}{3} \operatorname{Tr} \left(y_1 \ (y_1)^\dagger \right) + \frac{4}{3} \operatorname{Tr} \left(y_2 \ (y_2)^\dagger \right) \right) + \frac{4}{3} \lambda_{T3} \ (\lambda_{T3})^*$		

Terms 9 of 11

Column	Representative term structure
T1	$+16g_2^4$
T2	$+g_2^2(-24\lambda_{12} - 16\lambda_{12}^{\text{Cross}})$
T3	$+g_Y^2(-6\lambda_{12} - 4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16g_2^4$	$+g_2^2(-24\lambda_{12} - 16\lambda_{12}^{\text{Cross}})$	$+g_Y^2(-6\lambda_{12} - 4\lambda_{12}^{\text{Cross}})$

Terms 10 of 11

Column	Representative term structure
T1	$+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$
T2	$+\lambda_{S_2}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$
T3	$+16\lambda_{12}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$	$+\lambda_{S_2}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$	$+16\lambda_{12}^{\text{Adj}^2}$

Terms 11 of 11

Column	Representative term structure	
T1	$+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$	
T2	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$	
T3	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$

$\beta_{\lambda_{T3}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{T3}}{d\mu}=\beta_{\lambda_{T3}}^{(1)}$

Terms 1 of 6

Column		Representative term structure
T1	λ_{T3}	$\left(\text{Tr}\left(y_1\ (y_1)^\dagger\right)+\text{Tr}\left(y_2\ (y_2)^\dagger\right)+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$
T2	$+2$	$\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3	$+2$	$\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\ (y_2)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{T3}\left(\text{Tr}\left(g_1\ (g_1)^\dagger\right)+\text{Tr}\left(g_2\ (g_2)^\dagger\right)+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$	$+2\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$	$+2\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\ (y_2)^\dagger\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+3\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$	$+3\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\ (y_2)^\dagger\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 6

Column	Representative term structure
T1	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$
T2	$+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$
T3	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left(\lambda_{S_1^\dagger S_2 S_2^\dagger} \right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left(\lambda_{S_1^\dagger S_2 S_2^\dagger} \right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left(\lambda_{S_1^\dagger S_2 S_2^\dagger} \right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 6

Column	Representative term structure
T1	$-9g_2^2\lambda_{T3}$
T2	$-3g_Y^2\lambda_{T3}$
T3	$+2\lambda_{12}\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{T3}$	$-3g_Y^2\lambda_{T3}$	$+2\lambda_{12}\lambda_{T3}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{T3}$	$-3g_Y^2\lambda_{T3}$	$+2\lambda_{12}\lambda_{T3}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_{12}\lambda_{T3}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+2\lambda_{12}\lambda_{T3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_{12}\lambda_{T3}$

Terms 4 of 6

Column	Representative term structure
T1	$+2\lambda_H\lambda_{T3}$
T2	$+4\lambda_{H1}\lambda_{T3}$
T3	$+4\lambda_{H2}\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$	$+4\lambda_{H2}\lambda_{T3}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$	$+4\lambda_{H2}\lambda_{T3}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$	$+4\lambda_{H2}\lambda_{T3}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$	$+4\lambda_{H2}\lambda_{T3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$	$+4\lambda_{H2}\lambda_{T3}$

Terms 5 of 6

Column	Representative term structure	
T1	$\lambda_{T3}\left(\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{3}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$	
T2	$\lambda_{T3}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{H2}^{\text{Adj}})$	
T3	$-\frac{21g_2^2\lambda_{T3}}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{T3}\left(\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{3}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{T3}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{H2}^{\text{Adj}})$	$-\frac{21g_2^2\lambda_{T3}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$-\frac{21g_2^2\lambda_{T3}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$-\frac{33g_2^2\lambda_{T3}}{2}$

Terms 6 of 6

Column	Representative term structure
T1	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T2	$\lambda_{T3}(\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)-4\lambda_{H1}^{\text{Adj}})$
T3	$\lambda_{T3}\left(\text{Tr}\big(y_1\ (y_1)^\dagger\big)+\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)-\frac{4\lambda_{12}^{\text{Adj}}}{3}-2\lambda_{12}^{\text{Cross}}-2\lambda_{H1}^{\text{Adj}}-\frac{4\lambda_{H2}^{\text{Adj}}}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$	$\lambda_{T3}(\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)-4\lambda_{H1}^{\text{Adj}})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$	—	$\lambda_{T3}\left(\text{Tr}\big(y_1\ (y_1)^\dagger\big)+\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)-\frac{4\lambda_{12}^{\text{Adj}}}{3}-2\lambda_{12}^{\text{Cross}}-2\lambda_{H1}^{\text{Adj}}-\frac{4\lambda_{H2}^{\text{Adj}}}{3}\right)$

$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure
T1	$\lambda_{12}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$
T2	$+\lambda_{S_1^\dagger S_2 S_2^\dagger}2\left(\frac{2}{3}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+\frac{4}{3}(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$
T3	$+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\left(\frac{2}{3}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+\frac{4}{3}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}\right)^*+5\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$	$+\lambda_{S_1^\dagger S_2 S_2^\dagger}\left(\frac{2}{3}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+\frac{4}{3}(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$	$+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\left(\frac{2}{3}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+\frac{4}{3}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\right)^*+5\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{12}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+3\text{Tr}\big(y_2\ (y_2)^\dagger\big))$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 8

Column	Representative term structure	
T1	$+\frac{2}{3}((\lambda_{S_1^\dagger S_2 S_2^\dagger 2})^* + (\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})^*) \text{Tr}(y_2 (y_1)^\dagger)$	
T2	$-\frac{8}{3} \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$	
T3	$+\frac{64}{3} \lambda_{S_1^\dagger 2 S_2^\dagger 2} \left(\lambda_{S_1^\dagger 2 S_2^\dagger 2} \right)^*$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{2}{3}((\lambda_{S_1^\dagger S_2 S_2^\dagger 2})^* + (\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})^*) \text{Tr}(y_2 (y_1)^\dagger)$	$-\frac{8}{3} \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$	$+\frac{64}{3} \lambda_{S_1^\dagger 2 S_2^\dagger 2} \left(\lambda_{S_1^\dagger 2 S_2^\dagger 2} \right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{2}{3} \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$	$+\frac{64}{3} \lambda_{S_1^\dagger 2 S_2^\dagger 2} \left(\lambda_{S_1^\dagger 2 S_2^\dagger 2} \right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{2}{3} \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$	—

Terms 3 of 8

Column	Representative term structure
T1	$+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$
T2	$+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$
T3	$-9g_2^2\lambda_{12}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$	$-9g_2^2\lambda_{12}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$	$-9g_2^2\lambda_{12}^{\text{Adj}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2}{9}\lambda_{T3}\left(\lambda_{T3}\right)^*$	—	$-24g_2^2\lambda_{12}^{\text{Adj}}$

Terms 4 of 8

Column	Representative term structure
T1	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T2	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$
T3	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$

Terms 5 of 8

Column	Representative term structure
T1	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T2	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+\frac{4}{3}\left(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*\right)\right)$
T3	$+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\left(\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+\frac{4}{3}\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+5\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*\right)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+\frac{4}{3}\left(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*\right)\right)$	$+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\left(\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+\frac{4}{3}\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+5\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*\right)\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 6 of 8

Column	Representative term structure
T1	$\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right)$
T2	$-\frac{8}{3} \text{Tr} \left(y_1 \left(y_2 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$
T3	$\lambda_{12}^{\text{Adj}} \left(2 \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + 2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) - \frac{8}{3} \text{Tr} \left(y_1 \left(y_2 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\lambda_{12}^{\text{Adj}} \left(2 \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + 2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$

Terms 7 of 8

Column	Representative term structure
T1	$-6g_Y{}^2\lambda_{12}^{\text{Adj}}$
T2	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$
T3	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-6g_Y{}^2\lambda_{12}^{\text{Adj}}$	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$

Column	Representative term structure
T1	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

$\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Cross}}}{d\mu}=\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)$
T2	$+\lambda_{12}^{\text{Cross}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\right)$
T3	$-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)+\lambda_{12}^{\text{Cross}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\right)$	$-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$
T2	$+\frac{33g_2^4}{4}$
T3	$+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$

Terms 3 of 5

Column	Representative term structure
T1	$+g_Y{}^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$
T2	$+\lambda_{12}{}^2$
T3	$+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$	$+\lambda_{12}{}^2$	$+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$

Terms 4 of 5

Column	Representative term structure
T1	$+\lambda_{S_2}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$
T2	$+4\lambda_{12}^{\text{Adj}^2}$
T3	$+11\lambda_{12}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{S_2}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$	$+4\lambda_{12}^{\text{Adj}^2}$	$+11\lambda_{12}^{\text{Cross}^2}$

Terms 5 of 5

Column	Representative term structure
T1	$-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$
T2	$-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$
T3	$+\lambda_{H1}\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$	$-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$	$+\lambda_{H1}\lambda_{H2}$

$\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{H1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\frac{2}{3}\text{Tr}\big(y_1\ (y_2)^\dagger\big)-\frac{4}{3}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)\right)$
T3	$-\frac{2}{3}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\frac{2}{3}\text{Tr}\big(y_1\ (y_2)^\dagger\big)-\frac{4}{3}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)\right)$	$-\frac{2}{3}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 7

Column	Representative term structure
T1	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T2	$+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$
T3	$-\frac{4}{3}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$-\frac{4}{3}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$-\frac{4}{3}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	—	—

Terms 3 of 7

Column	Representative term structure
T1	$+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$
T2	$+g_Y^2\left(-6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$
T3	$-9g_2^2\lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(-6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$	$-9g_2^2\lambda_{H1}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(-6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$	$-9g_2^2\lambda_{H1}^{\text{Adj}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	—	—

Terms 4 of 7

Column	Representative term structure
T1	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T2	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$
T3	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$

Terms 5 of 7

Column	Representative term structure
T1	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S1S_1^{\dagger 2}S_2^\dagger}\right)^*\right)\right)$
T3	$-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S1S_1^{\dagger 2}S_2^\dagger}\right)^*\right)\right)$	$-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	—	—

Terms 6 of 7

Column	Representative term structure
T1	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\ (y_1)^{\dagger}\right)+6\text{Tr}\left(Y_d\ (Y_d)^{\dagger}\right)+2\text{Tr}\left(Y_e\ (Y_e)^{\dagger}\right)+6\text{Tr}\left(Y_u\ (Y_u)^{\dagger}\right)+16\lambda_{S1}^{\text{Adj}})$
T2	$+g_Y^2\left(-4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$
T3	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\ (y_1)^{\dagger}\right)+6\text{Tr}\left(Y_d\ (Y_d)^{\dagger}\right)+2\text{Tr}\left(Y_e\ (Y_e)^{\dagger}\right)+6\text{Tr}\left(Y_u\ (Y_u)^{\dagger}\right)+16\lambda_{S1}^{\text{Adj}})$	$+g_Y^2\left(-4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\ (y_1)^{\dagger}\right)+6\text{Tr}\left(Y_d\ (Y_d)^{\dagger}\right)+2\text{Tr}\left(Y_e\ (Y_e)^{\dagger}\right)+6\text{Tr}\left(Y_u\ (Y_u)^{\dagger}\right)+16\lambda_{S1}^{\text{Adj}})$	—	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$

Terms 7 of 7

Column	Representative term structure
T1	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$
T2	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

$\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{H2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$

Terms 1 of 7

Column	Representative term	structure
T1	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2\ (y_2)^\dagger)+6\text{Tr}(Y_d\ (Y_d)^\dagger)+2\text{Tr}(Y_e\ (Y_e)^\dagger)+6\text{Tr}(Y_u\ (Y_u)^\dagger))$	
T2	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-2\text{Tr}(y_1\ (y_2)^\dagger)-4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*))$	
T3	$-2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}(y_2\ (y_1)^\dagger)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2\ (y_2)^\dagger)+6\text{Tr}(Y_d\ (Y_d)^\dagger)+2\text{Tr}(Y_e\ (Y_e)^\dagger)+6\text{Tr}(Y_u\ (Y_u)^\dagger))$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-2\text{Tr}(y_1\ (y_2)^\dagger)-4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*))$	$-2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}(y_2\ (y_1)^\dagger)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H2}^{\text{Adj}}(3\text{Tr}(y_2\ (y_2)^\dagger)+6\text{Tr}(Y_d\ (Y_d)^\dagger)+2\text{Tr}(Y_e\ (Y_e)^\dagger)+6\text{Tr}(Y_u\ (Y_u)^\dagger))$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-3\text{Tr}(y_1\ (y_2)^\dagger)-4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*))$	$-3\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}(y_2\ (y_1)^\dagger)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 7

Column	Representative term structure
T1	$+8\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$
T2	$-4\lambda_{S_1^{\dagger}S_2S_2^{\dagger}2}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$
T3	$+64\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$-4\lambda_{S_1^{\dagger}S_2S_2^{\dagger}2}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$	$+64\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$-4\lambda_{S_1^{\dagger}S_2S_2^{\dagger}2}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$	$+64\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 7

Column	Representative term structure
T1	$+16\lambda_{T3} \left(\lambda_{T3}\right)^*$
T2	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H2}^{\text{Adj}}\right)$
T3	$-9g_2^2 \lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+16\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H2}^{\text{Adj}}\right)$	$-9g_2^2 \lambda_{H2}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+16\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H2}^{\text{Adj}}\right)$	$-9g_2^2 \lambda_{H2}^{\text{Adj}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 7

Column	Representative term structure
T1	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T2	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$

Terms 5 of 7

Column	Representative term structure
T1	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$
T2	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S2}^{\text{Adj}})$
T3	$+\frac{4}{3}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$

Terms 6 of 7

Column	Representative term structure
T1	$+\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$
T2	$+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$
T3	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{8}{9}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$

Terms 7 of 7

Column	Representative term structure
T1	$+12\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$

$$\beta_{\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+(\lambda_{T3})^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+(\lambda_{T3})^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	—	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$+2\lambda_{S_1 S_1^\dagger{}^2 S_2^\dagger}(\lambda_{T3})^*$
T2	$-9g_2{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3	$-3g_Y{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{S_1 S_1^\dagger{}^2 S_2^\dagger}(\lambda_{T3})^*$	$-9g_2{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$-3g_Y{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{S_1 S_1^\dagger{}^2 S_2^\dagger}(\lambda_{T3})^*$	$-9g_2{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$-3g_Y{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T2	$+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3	$+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{S_1}$
T2	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$
T3	$+\frac{3}{2}(\lambda_{T3})^* \text{Tr}\left(y_2 \left(y_1\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{S_1}$	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{S_1}$	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	$+\frac{3}{2}(\lambda_{T3})^* \text{Tr}\left(y_2 \left(y_1\right)^\dagger\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$$\beta_{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{d\mu}=\beta_{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure	
T1	$\lambda_{H1}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)$	
T2	$+\lambda_{H1}^{\text{Adj}}\left(-\frac{1}{8}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$	
T3	$+\lambda_{H2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H1}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)+\lambda_{H1}^{\text{Adj}}\left(-\frac{1}{8}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)+\lambda_{H2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)$		
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H1}\left(\frac{3}{4}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)+\lambda_{H1}^{\text{Adj}}\left(-\frac{3}{16}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)+\lambda_{H2}\left(\frac{3}{4}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)$		
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column		Representative term structure
T1	$+\lambda_{H2}^{\text{Adj}}\left(-\frac{3}{8}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)-\frac{3\lambda^{\text{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}}{2}-\frac{3\lambda_{S_1^{\dagger}S_2S_2^{\dagger}2}}{4}\right)$	
T2	$+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}(\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	
T3	$-4\text{Tr}\left(y_2\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{H2}^{\text{Adj}}\left(-\frac{3}{8}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)-\frac{3\lambda^{\text{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}}{2}-\frac{3\lambda_{S_1^{\dagger}S_2S_2^{\dagger}2}}{4}\right)$	$+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}(\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$-4\text{Tr}\left(y_2\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{H2}^{\text{Adj}}\left(-\frac{9}{16}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)-\frac{3\lambda^{\text{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}}{2}-\frac{3\lambda_{S_1^{\dagger}S_2S_2^{\dagger}2}}{4}\right)$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$+6\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T2	$-6 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$
T3	$-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$-6 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$	$-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$-6 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$	$-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure	
T1	$-3g_Y{}^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	
T2	$+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	
T3	$+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_Y{}^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_Y{}^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}{}^{\lambda_{T3}}$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\frac{3}{2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}{}^{\lambda_{T3}}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}{}^{\lambda_{T3}}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{2}{3}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\frac{3}{2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{H^\dagger H^\dagger S_2 S_2}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H^\dagger H^\dagger S_2 S_2}(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T2	$+\left(\lambda_{T3}\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$
T3	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H^\dagger H^\dagger S_2 S_2}(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+\left(\lambda_{T3}\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H^\dagger H^\dagger S_2 S_2}(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	—	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$
T2	$-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	$-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	$-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$
T2	$+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3	$+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_2}$
T2	$+\frac{3}{2}(\lambda_{T3})^* \text{Tr}\Big(y_1 \ (y_2)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_2}$	$+\frac{3}{2}(\lambda_{T3})^* \text{Tr}\Big(y_1 \ (y_2)^\dagger\Big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$
T2	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$
T3	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$

Terms 2 of 6

Column	Representative term structure
T1	$+ \frac{3g_2^4}{4}$
T2	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$
T3	$+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$	$+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$

Terms 3 of 6

Column	Representative term structure
T1	$+\frac{3g_Y^4}{2}$
T2	$+\frac{\lambda_{12}^2}{4}$
T3	$+\frac{\lambda_{H1}^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3g_Y^4}{2}$	$+\frac{\lambda_{12}^2}{4}$	$+\frac{\lambda_{H1}^2}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+\frac{\lambda_{H1}^2}{2}$

Terms 4 of 6

Column	Representative term structure
T1	$+3\lambda_{H1}^{\text{Adj}^2}$
T2	$+\frac{7\lambda_{S1}^2}{4}$
T3	$+22\lambda_{S1}^{\text{Adj}^2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}^2}$

Terms 5 of 6

Column	Representative term structure
T1	$+\frac{3\lambda_{12}^2}{4}$
T2	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3	$+12\lambda_{12}^{\text{Adj}2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+12\lambda_{12}^{\text{Adj}2}$

Terms 6 of 6

Column		Representative term structure
T1	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$

$\beta_{\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}{d\mu}=\beta_{\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}(\text{Tr}\left(y_2\ (y_1)^{\dagger}\right)+6\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}+6\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}})$
T2	$+\lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\left(y_2\ (y_1)^{\dagger}\right)+\frac{\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}}{6}+\frac{5\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}}{6}\right)$
T3	$+\lambda_{S_1}(\text{Tr}\left(y_2\ (y_1)^{\dagger}\right)+12\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}(\text{Tr}\left(y_2\ (y_1)^{\dagger}\right)+6\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}+6\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}})$	$+\lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\left(y_2\ (y_1)^{\dagger}\right)+\frac{\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}}{6}+\frac{5\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}}{6}\right)$	$+\lambda_{S_1}(\text{Tr}\left(y_2\ (y_1)^{\dagger}\right)+12\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\lambda_{12}^{\text{Adj}}\left(\frac{1}{8}\text{Tr}\left(y_2\ (y_1)^{\dagger}\right)+\frac{\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}}{6}+\frac{5\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}}{6}\right)$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure	
T1	$+ \lambda_{S_1^\dagger 2 S_2^\dagger 2} (2 \operatorname{Tr} (y_1 \ (y_2)^\dagger) + 4 (\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*))$	
T2	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} (3 \operatorname{Tr} (y_1 \ (y_1)^\dagger) + \operatorname{Tr} (y_2 \ (y_2)^\dagger))$	
T3	$- 4 \operatorname{Tr} (y_1 \ (y_1)^\dagger \ y_2 \ (y_1)^\dagger)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_{S_1^\dagger 2 S_2^\dagger 2} (2 \operatorname{Tr} (y_1 \ (y_2)^\dagger) + 4 (\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*))$	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} (3 \operatorname{Tr} (y_1 \ (y_1)^\dagger) + \operatorname{Tr} (y_2 \ (y_2)^\dagger))$	$- 4 \operatorname{Tr} (y_1 \ (y_1)^\dagger \ y_2 \ (y_1)^\dagger)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_1^\dagger 2 S_2^\dagger 2} (3 \operatorname{Tr} (y_1 \ (y_2)^\dagger) + 4 (\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*))$	—	$- 5 \operatorname{Tr} (y_1 \ (y_1)^\dagger \ y_2 \ (y_1)^\dagger)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$-9g_2^2\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}$
T2	$-3g_Y^2\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}$
T3	$+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}$	$-3g_Y^2\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}$	$+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}$	$-3g_Y^2\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}$	$+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$
T2	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$
T3	$\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$	$\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+ \lambda_{S_1} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)$
T2	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\frac{9}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{3}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_1} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)$	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\frac{9}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{3}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}{d\mu}=\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(2\text{Tr}\big(y_1\ (y_1)^{\dagger}\big)+2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big))$
T2	$+ \lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+2\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}})$
T3	$+ \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(2\text{Tr}\big(y_1\ (y_1)^{\dagger}\big)+2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big))$	$+ \lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+2\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}})$	$+ \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(3\text{Tr}\big(y_1\ (y_1)^{\dagger}\big)+3\text{Tr}\big(y_2\ (y_2)^{\dagger}\big))$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 6

Column	Representative term structure
T1	$-2\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$
T2	$+4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^{*}$
T3	$-9g_2^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-2\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^{*}$	$-9g_2^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^{*}$	$-9g_2^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 6

Column	Representative term structure
T1	$-3g_Y{}^2\lambda_{S_1^\dagger 2S_2^\dagger 2}$
T2	$+8\lambda_{12}\lambda_{S_1^\dagger 2S_2^\dagger 2}$
T3	$+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^\dagger 2S_2^\dagger 2}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_Y{}^2\lambda_{S_1^\dagger 2S_2^\dagger 2}$	$+8\lambda_{12}\lambda_{S_1^\dagger 2S_2^\dagger 2}$	$+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^\dagger 2S_2^\dagger 2}}{3}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_Y{}^2\lambda_{S_1^\dagger 2S_2^\dagger 2}$	$+8\lambda_{12}\lambda_{S_1^\dagger 2S_2^\dagger 2}$	$+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^\dagger 2S_2^\dagger 2}}{3}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 6

Column	Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$
T2	$+2\lambda_{S_1} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$
T3	$+2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$	$+2\lambda_{S_1} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	$+2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \lambda_{S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$	$+2\lambda_{S_1} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	$+2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \lambda_{S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 6

Column	Representative term structure
T1	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$
T2	$+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$
T3	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 2\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$	$+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$	$+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 2\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 6 of 6

Column	Representative term structure
T1	$+\frac{3}{2}\lambda_{S_1S_1^\dagger 2S_2^\dagger}\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$
T2	$-\frac{5}{2}\text{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}\lambda_{S_1S_1^\dagger 2S_2^\dagger}\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$	$-\frac{5}{2}\text{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}^{(1)}$

$$16\pi^2\,\mu\frac{d\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{d\mu}=\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}(\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger^2}+6\lambda_{S_1 S_1^\dagger^2 S_2^\dagger})$
T2	$+\lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6}+\frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)$
T3	$+\lambda_{S_1^\dagger^2 S_2^\dagger^2}(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger^2}\right)^*+\left(\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}\right)^*))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}(\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger^2}+6\lambda_{S_1 S_1^\dagger^2 S_2^\dagger})$	$+\lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6}+\frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)$	$+\lambda_{S_1^\dagger^2 S_2^\dagger^2}(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger^2}\right)^*+\left(\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}\right)^*))$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\lambda_{12}^{\text{Adj}}\left(\frac{1}{8}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6}+\frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)$	$+\lambda_{S_1^\dagger^2 S_2^\dagger^2}(3\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger^2}\right)^*+\left(\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}\right)^*))$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$+ \lambda_{S_2} (\text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{S_1^\dagger S_2 S_2^\dagger 2})$
T2	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} (\text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + 3 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right))$
T3	$- 4 \text{Tr} \left(y_2 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_{S_2} (\text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{S_1^\dagger S_2 S_2^\dagger 2})$	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} (\text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + 3 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right))$	$- 4 \text{Tr} \left(y_2 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$- 5 \text{Tr} \left(y_2 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$
T2	$-9 g_2^2 \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$
T3	$-3 g_Y^2 \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$	$-9 g_2^2 \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$	$-3 g_Y^2 \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$	$-9 g_2^2 \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$	$-3 g_Y^2 \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$
T3	$\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+ \lambda_{S_2} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)$
T2	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{3}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{9}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_2} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)$	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{3}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{9}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$
T2	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$
T3	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Terms 2 of 6

Column	Representative term structure
T1	$+ \frac{3g_2^4}{4}$
T2	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_2}}{2} - 12\lambda_{S_2}^{\text{Adj}} \right)$
T3	$+g_2^2 (-3\lambda_{S_2} - 24\lambda_{S_2}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_2^4}{4}$	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_2}}{2} - 12\lambda_{S_2}^{\text{Adj}} \right)$	$+g_2^2 (-3\lambda_{S_2} - 24\lambda_{S_2}^{\text{Adj}})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_2^4}{4}$	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_2}}{2} - 12\lambda_{S_2}^{\text{Adj}} \right)$	$+g_2^2 (-3\lambda_{S_2} - 24\lambda_{S_2}^{\text{Adj}})$

Terms 3 of 6

Column	Representative term structure
T1	$+\frac{3g_Y^4}{2}$
T2	$+\frac{\lambda_{12}^2}{4}$
T3	$+\frac{\lambda_{H2}^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_Y^4}{2}$	$+\frac{\lambda_{12}^2}{4}$	$+\frac{\lambda_{H2}^2}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_Y^4}{2}$	—	$+\frac{\lambda_{H2}^2}{2}$

Terms 4 of 6

Column	Representative term structure
T1	$+3\lambda_{H2}^{\text{Adj}^2}$
T2	$+\frac{7\lambda_{S2}^2}{4}$
T3	$+22\lambda_{S2}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$	$+22\lambda_{S2}^{\text{Adj}^2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$	$+22\lambda_{S2}^{\text{Adj}^2}$

Terms 5 of 6

Column	Representative term structure
T1	$+\frac{3\lambda_{12}^2}{4}$
T2	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3	$+12\lambda_{12}^{\text{Adj}2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+12\lambda_{12}^{\text{Adj}2}$

Terms 6 of 6

Column	Representative term structure
T1	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$